DIAZEPAM

Controlled drug
Respiratory depression

CHEMICAL NAME 7-chloro-1-methyl-5-phenyl-1,3-dihydro-2H-1,4-benzodiazepin-2-one
CLASS Long-acting benzodiazepine
TRADE NAMES Valium · Stesolid (rectal) · Diazemuls · Valtoco (nasal)

THERAPEUTIC USES / INDICATIONS

- Status epilepticus / prolonged seizures (IV, rectal, buccal alternative).
- Alcohol withdrawal.
- Acute anxiety (short-term); muscle spasm; tetanus.
- Febrile convulsions (rectal); pre-procedure sedation.

ADMINISTRATION

- **IV** slowly into large vein (max 5 mg/min) — emulsion (Diazemuls) less irritant.
- **Rectal** tubes (child-friendly).
- **PO** tablets/solution.
- Avoid IM (erratic absorption, painful).

DOSAGES

- **Status epilepticus (adult):** 10 mg IV (or 10–20 mg PR), repeat once after 10 min.
- **Children:** 0.3–0.4 mg/kg IV (max 10 mg); PR 0.5 mg/kg (max 10–20 mg).
- **Anxiety:** 2–10 mg 2–3 times daily (max 4 weeks).
- **Alcohol withdrawal:** 10–20 mg four times daily, tapered (or symptom-triggered).
- **Elderly:** half dose.

PHARMACOKINETICS

- **Onset:** IV 1–3 min; PR 5–10 min; PO 30–60 min. **Duration:** IV anticonvulsant 15–30 min (redistribution); sedation long.
- Metabolised in liver (CYP3A4, 2C19) to active desmethyldiazepam (t½ up to 100 h).
- Excreted in urine. **t½:** 20–50 h (longer in elderly).

PHARMACODYNAMICS

- Positive allosteric modulator of **GABA-A receptors** → ↑ frequency of chloride channel opening → anxiolytic, sedative, anticonvulsant, muscle relaxant, amnesic.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Drowsiness, sedation, fatigue.
- Ataxia, dizziness, confusion (elderly).
- Anterograde amnesia.

ADVERSE EFFECTS (SERIOUS)

- Respiratory depression (with opioids, alcohol, elderly, COPD), hypotension (IV).
- Dependence, tolerance, withdrawal (seizures, delirium).
- Paradoxical agitation; falls and fractures in elderly.
- Thrombophlebitis (IV).

CONTRAINDICATIONS & CAUTIONS

- Respiratory depression; severe respiratory insufficiency; sleep apnoea.
- Myasthenia gravis.
- Severe hepatic impairment.
- Acute pulmonary insufficiency; phobic/obsessional states alone.

KEY INTERACTIONS

- **Opioids, alcohol, other sedatives:** profound sedation, death.
- **CYP3A4/2C19 inhibitors (fluconazole, omeprazole):** ↑ levels.
- **Phenytoin:** variable.

NURSING MANAGEMENT

- Monitor RR, SpO_2, BP, consciousness; keep airway equipment, oxygen and flumazenil available.
- Time seizures; give second dose only after 10 min if still seizing.
- Falls risk assessment.
- Short courses; taper long-term users.

PATIENT EDUCATION

- Do not drive or drink alcohol.
- Do not stop suddenly after regular use.
- Rectal diazepam: teach family for prolonged seizures (>5 min).
- Risk of dependence — short-term use only.

KEY POINTS

- Long-acting benzodiazepine (active metabolite).
- Respiratory depression; antidote flumazenil.
- Status: 10 mg IV (max 5 mg/min) or PR.
- Dependence; avoid IM.